课程笔记 SNN训练算法汇总综述

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摘要

课程笔记 SNN训练算法汇总综述，涵盖了从传统的 BPTT/STBP 到最新的在线学习方法（OTTT、NDOT）以及局部学习规则（S-TLLR、TESS）。重点分析了每种方法的核心思想、数学推导、优缺点以及适用场景。通过对比不同算法在时间依赖性处理、梯度计算复杂度和生物可解释性方面的差异，全面总结了 SNN 训练领域的研究进展和未来方向。

PPT $Week4\_SNNTraining.pptx$

文章汇总

RTRL: **A learning algorithm for continually running fully recurrent neural networks** [1];

E-prop: **A solution to the learning dilemma for recurrent networks of spiking neurons** [2];

OTTT: **Online Training Through Time for Spiking Neural Networks** [3];

NDOT: **NDOT: Neuronal Dynamics-based Online Training for Spiking Neural Networks** [4];

S-TLLR: **S-TLLR: STDP-inspired Temporal Local Learning Rule for Spiking Neural Networks** [5];

TESS: **TESS: A Scalable Temporally and Spatially Local Learning Rule for Spiking Neural Networks** [6]

参考文献

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